

```

prepare("SELECT status FROM document_meta WHERE doc_id = ?"); $meta_stmt-
>execute(array($doc_id)); $status_val = $meta_stmt->fetchColumn(); if ($status_val ===
'FINALIZED_AND_LOCKED') { $is_locked = true; } } catch (Exception $e) {} } $authenticated_user =
isset($_SESSION['authenticated_user']) ? $_SESSION['authenticated_user'] :
(isset($_COOKIE['sands_auth_device']) ? $_COOKIE['sands_auth_device'] : ''); // Handle Ajax Actions if
(isset($_SERVER['REQUEST_METHOD']) && $_SERVER['REQUEST_METHOD'] === 'POST' &&
isset($_POST['action'])) { header('Content-Type: application/json'); $action = $_POST['action']; if
(!$authenticated_user) { echo json_encode(array('success' => false, 'message' => 'Unauthorized access.
Please log in via the portal.)); exit; } if ($action === 'sign_milestone') { $role = isset($_POST['role']) ?
trim($_POST['role']) : ''; $signer_name = isset($_POST['signer_name']) ? trim($_POST['signer_name']) : '';
$signer_designation = isset($_POST['signer_designation']) ? trim($_POST['signer_designation']) : '';
$signature_data = isset($_POST['signature_data']) ? trim($_POST['signature_data']) : ''; if (empty($role) ||
empty($signer_name) || empty($signature_data)) { echo json_encode(array('success' => false, 'message'
=> 'Missing required signature fields.)); exit; } try { $now = date('Y-m-d H:i:s'); // Check if document record
exists in document_meta $check_stmt = $pdo->prepare("SELECT COUNT(*) FROM document_meta
WHERE doc_id = ?"); $check_stmt->execute(array($doc_id)); if ($check_stmt->fetchColumn() == 0) {
$ins_stmt = $pdo->prepare("INSERT INTO document_meta (doc_id, title, status) VALUES (?, ?,
'IN_REVIEW')"); $ins_stmt->execute(array($doc_id, 'Module 8: Hardware Integration, QR Handheld
Devices, Thermal Printers & Server Setup Infrastructure Milestone')); } if ($db_driver_active === 'mysql') {
$sig_stmt = $pdo->prepare("INSERT INTO document_signatures (doc_id, role, signer_name,
signer_designation, signer_email, signature_data, signed_at) VALUES (?, ?, ?, ?, ?, ?, ?) ON DUPLICATE KEY
UPDATE signer_name = VALUES(signer_name), signer_designation = VALUES(signer_designation),
signature_data = VALUES(signature_data), signed_at = VALUES(signed_at)"); $sig_stmt-
>execute(array($doc_id, $role, $signer_name, $signer_designation, $authenticated_user, $signature_data,
$now)); } else { $del = $pdo->prepare("DELETE FROM document_signatures WHERE doc_id = ? AND role =
?"); $del->execute(array($doc_id, $role)); $ins = $pdo->prepare("INSERT INTO document_signatures
(doc_id, role, signer_name, signer_designation, signer_email, signature_data, signed_at) VALUES (?, ?, ?, ?,
?, ?, ?)"); $ins->execute(array($doc_id, $role, $signer_name, $signer_designation, $authenticated_user,
$signature_data, $now)); } // Log to audit trail $audit_stmt = $pdo->prepare("INSERT INTO audit_trail
(user_email, action, details, ip_address) VALUES (?, ?, ?, ?)"); $audit_stmt-
>execute(array($authenticated_user, 'SIGN_DOCUMENT', "Signed $doc_id as $role ($signer_name)",
$_SERVER['REMOTE_ADDR'])); echo json_encode(array('success' => true, 'message' => 'Signature
recorded successfully!', 'signed_at' => $now)); exit; } catch (Exception $e) { echo
json_encode(array('success' => false, 'message' => 'Database error: ' . $e->getMessage())); exit; } } if
($action === 'get_signatures') { try { $stmt = $pdo->prepare("SELECT role, signer_name,
signer_designation, signer_email, signature_data, signed_at FROM document_signatures WHERE doc_id =
?"); $stmt->execute(array($doc_id)); $signatures = $stmt->fetchAll(PDO::FETCH_ASSOC); echo
json_encode(array('success' => true, 'signatures' => $signatures)); exit; } catch (Exception $e) { echo
json_encode(array('success' => false, 'message' => 'Error fetching signatures: ' . $e->getMessage())); exit;
} } echo json_encode(array('success' => false, 'message' => 'Unknown action.)); exit; } ?>

```

Module 8: Hardware Integration, QR Code Handheld Device Integration, Biometric Clocks, ESC/POS Printers & Server Infrastructure Setup

Comprehensive 16-week milestone agreement, dedicated engineering team breakdown, Bahrain regulatory compliance matrix (SIO / GOSI / LMRA / WPS), and payment schedule based on the verified Business Analysis Report **ARCH-001 v1.0** for **Popular Auto Spare & A/C Parts Co. W.L.L.**

DOCUMENT ID	BA REFERENCE DOCUMENT	TOTAL ENGINEERING EFFORT
SL-POP-ERP-MS-008	ARCH-001 v1.0 (04-July-2026)	3 Working Weeks (0.75 Months / 15 Days)
FIXED PROJECT COST	TARGET ARCHITECTURE	DOCUMENT STATUS
BD 1,022.727	3-Tier Server Cluster, QR Handhelds, Thermal Printers & Biometrics	FINAL PROPOSAL

Executive Summary & Master ERP Context

Hardware Integration & Server Infrastructure (Module 8) establishes the foundational compute, networking, cybersecurity, hardware device connectivity, and offline synchronization topology powering the entire ERP transformation for **Popular Auto Spare & A/C Parts Co. W.L.L.** Derived from the master **ERP Technical Architecture & Cybersecurity Specification (ARCH-001 v1.0)**, this module delivers high-availability 3-tier domain isolation (Development, Staging/Sandbox, Production), RabbitMQ asynchronous messaging daemons, offline-first SQLite bi-directional synchronization, Android handheld QR code barcode scanner integration, ESC/POS thermal receipt and label printing daemons, RJ11 cash drawer triggers, and ZKTeco/Hikvision biometric time-clock listeners across all branch showrooms and warehouses.



Core Objective & Business Purpose

To commission a resilient enterprise infrastructure and hardware integration suite: 3-Tier Isolated Cloud Server Environments (Dev/Staging/Prod), RabbitMQ Message Queue Service, JetBackup 5 Daily Backups, Handheld Android 1D/2D QR Barcode Scanner Listeners, ESC/POS Thermal Receipt & Barcode Printers, RJ11 Cash Drawer Electronic Triggers, Biometric Attendance Hardware Daemons, and Zero-Trust AES-256 Encrypted Networks.

Master ERP Transformation Architecture (9 Core Process Modules)

CURRENT SCOPE: MODULE 08
ACTIVE

- 01 PCode Gen & Item Master
- 02 Vendor & Purchase Flow
- 03 Stock Verification & Location
- 04 Sales & POS Checkout
- 05 Finance, Tax & VAT Accounting
- 06 Administration & Security

07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure

09 Executive BI & Mobile Analytics



3-Tier Server Cluster & RabbitMQ

Complete multi-environment deployment (dev.popular.com, sandbox.popular.com, erp.popular.com) with RabbitMQ message broker, Redis memory caching, and automated JetBackup 5.

3-Tier Cluster

RabbitMQ Broker

SSL / TLS 1.3



QR Handheld & ESC/POS Printers

Zebra / Honeywell / Sunmi Android mobile terminal listeners, camera QR scanners, ESC/POS thermal receipt printers, EPL/ZPL thermal barcode label printers, and RJ11 cash drawers.

Android Handhelds

QR Listeners

ESC/POS Printers



Offline Auto-Sync & Biometrics

Offline-first local SQLite/IndexedDB queue with bi-directional auto-sync, TCP/IP ZKTECO biometric clock daemons, and zero-trust AES-256 encryption.

Offline Sync

Biometric TCP/IP

AES-256 Security

Module 6: Functional Scope & Architecture Breakdown (BA Ref: ARCH-001)

Derived directly from the 25 pages of **ARCH-001: Administration Modules Business Analysis Report**, the implementation scope is structured across 4 core operational pillars:

2.1 Facility & Workplace Office Management

- **Facility Master:** Centralized registry of corporate offices, regional branches, central warehouses, workshops, retail outlets, and staff accommodations.
- **Maintenance Work Orders:** Preventive scheduled maintenance calendars, AMC vendor contracts, breakdown ticketing, and repair sign-off.
- **Employee Service Requests:** Internal ticketing portal for electrical, HVAC/air conditioning, plumbing, carpentry, and IT infrastructure requests.
- **Utility & Consumption Control:** Monthly tracking of electricity, water, internet, telephone, gas, and generator fuel with budget variance analysis.
- **Visitor Management (VMS):** Digital visitor registration, badge generation, host employee notification, and entry/exit timestamp logging.
- **Lease & Property Governance:** Landlord agreements, rent payment schedules, commercial lease terms, and automated renewal notifications.

2.2 Fixed Asset Management & QR Lifecycle Tracking

- **Asset Master Registry:** Comprehensive cataloging of IT equipment, warehouse machinery, tools, office furniture, vehicles, and facilities.
- **QR Code Tagging & Scanner:** Unique asset QR label generation with handheld scanner integration for rapid location identification and audits.
- **Asset Custody & Movements:** Tracking of employee custodians, department assignments, and inter-branch transfer delivery challans.
- **Warranty & AMC Management:** Supplier warranty coverage tracking, AMC renewal alerts, and maintenance history logging.
- **Depreciation Engine:** Automated straight-line and reducing-balance depreciation posting integrated directly with General Ledger (GL).

- **Physical Audit & Disposal:** Periodic physical inventory verification, variance logging, asset write-off, and scrap disposal workflows.

2.3 Fleet Logistics, Vehicle Maintenance & Total Cost of Ownership

- **Vehicle Master Registry:** Profile management for commercial delivery vans, heavy transport trucks, pickups, and corporate cars.
- **Vehicle Allocation & Booking:** Driver assignment, delivery route scheduling, key management, and trip odometer logging.
- **GPS & Route Monitoring:** Integration with telematics for real-time delivery tracking and mileage verification.
- **Preventive Maintenance:** Service intervals by mileage/date, oil changes, tire replacement logs, and workshop repair history.
- **Fuel Management:** Fuel card reconciliation, fuel receipt uploads, kilometer-per-liter efficiency analysis, and fuel theft anomaly detection.
- **Statutory Vehicle Compliance:** Traffic registration renewals, comprehensive insurance policies, fitness certificates, and accident claim workflows.
- **Vehicle TCO Analytics:** Real-time Total Cost of Ownership (TCO) tracking per vehicle combining fuel, maintenance, insurance, and depreciation.

2.4 Staff Locker & Corporate Document Management System

- **Staff Locker Management:** Branch physical locker registry, employee key allocation, locker inspections, and surrender protocols.
- **Digital Document Repository:** Encrypted digital vault for Commercial Registrations (CR), municipality licenses, lease agreements, vehicle titles, and contracts.
- **Role-Based Access Control (RBAC):** Granular permissions restricting document access to authorized executive and HR/Admin personnel.
- **Physical Document Movement:** Barcoded checkout/check-in tracking for original legal deeds, bank guarantees, and official certificates.
- **Automated Expiry & Renewal Alerts:** Proactive 90/60/30-day notifications for CR renewals, chamber of commerce, civil defense, and trade permits.
- **Compliance Audit Readiness:** Full historical audit trail of document uploads, version updates, downloads, and custodian modifications.

Project Team Structure & Dedicated Resource Pricing (12 Weeks / 3.0 Months)

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 12-week (3.0 months / 60 working days) delivery of Module 6 is structured as follows (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	3.0 Months (12 Wks)	BD 1,636.365
Back-End Lead Engineer (PHP MVC / REST APIs)	BD 227.273	BD 56.818	BD 1.420	3.0 Months (12 Wks)	BD 681.819
Front-End Lead Engineer (Admin UI / React Dashboards)	BD 227.273	BD 56.818	BD 1.420	3.0 Months (12 Wks)	BD 681.819
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	3.0 Months (12 Wks)	BD 613.635
QA & System Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	3.0 Months (12 Wks)	BD 477.273
Total Dedicated Engineering Team	BD 1,363.636 / mo	BD 340.909 / wk	BD 8.523 / hr	12 Weeks (3.0 Mo)	BD 1,022.727



Fixed Price Guarantee

Total project investment for Module 6 is fixed at **BD 1,022.727**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

Detailed Milestone Breakdown & Deliverables (12-Week Roadmap)

The implementation is organized into 4 sequential 3-week milestones (25.00% each), guaranteeing focused execution, continuous integration, and transparent deliverable acceptance:

1

Milestone 1: Facility & Workplace Management, Work Orders & Utility Control

WEEKS 1 – 3 (3 WKS)

Setup of centralized Facility Master, preventive/corrective maintenance work orders, employee complaint ticketing, utility consumption logs, visitor registration (VMS), and commercial property lease agreement renewals.



Facility Master & Zones

Centralized repository of all corporate branches, warehouses, and accommodation units.



Work Orders & Ticketing

Maintenance calendar, AMC vendor contracts, and employee service request portal.



Utility & Lease Control

Electricity/water cost tracking, budget variance, and automated lease renewal notifications.

2

Milestone 2: Fixed Asset Master, QR Tagging, Custodian Tracking & Depreciation

WEEKS 4 – 6 (3 WKS)

Engineering of Fixed Asset Master Registry, QR label generation, custodian assignments, inter-branch asset transfers, straight-line/reducing depreciation calculation, and physical audit verification tools.

3

✓ **QR Code Asset Tagging**

Unique QR label generation with handheld scanner barcode integration.

✓ **Asset Custody & Transfers**

Inter-branch transfer tracking, custodian sign-off, and movement history.

✓ **Depreciation & Audit Engine**

Automated monthly depreciation GL postings, warranty tracking, and scrap disposal.

Milestone 3: Fleet Logistics, Vehicle Bookings, Fuel Logs & TCO Analytics

WEEKS 7 – 9 (3 WKS)

Development of Vehicle Master Registry, driver allocation, GPS tracking integration, fuel consumption and card reconciliation, tire/parts service logs, insurance/road tax renewals, and total vehicle TCO analytics.

✓ **Vehicle Master & Bookings**

Fleet vehicle profiles, driver assignments, delivery trips, and key tracking.

✓ **Fuel & Mileage Analysis**

Fuel receipts, mileage efficiency (km/L), and fuel anomaly detection.

✓ **Compliance & TCO Dashboard**

Insurance/registration renewal alerts, maintenance logs, and vehicle TCO cockpit.

4

Milestone 4: Staff Locker Control, Secure Digital Document Vault & Final UAT

WEEKS 10 – 12 (3 WKS)

Implementation of Physical Locker Management, encrypted Digital Document Repository, physical document barcode movement tracking, automated statutory license renewal alerts (CR, Municipality), and multi-branch UAT sign-off.

✓ **Staff Locker Registry**

Branch locker allocation, key handovers, and locker inspection audits.

✓ **Secure Digital Document Vault**

Encrypted repository for trade licenses, CRs, title deeds, with RBAC security.

✓ **Statutory Renewal Engine & UAT**

Automated CR/permit renewal reminders, document movement tracking, and final UAT.

Milestone Payment Structure & Invoicing Schedule (12-Week Model)

Payments are structured in 4 equal milestone disbursements of **25.00% (BD 340.909 each)**, tied strictly to formal milestone deliverables and client sign-off:

Milestone Phase	Deliverable Description	Timeline	Percentage	Amount (BHD)	Payment Trigger & Acceptance Criteria
Milestone 1	Facility & Workplace Management, Work Orders & Utility Control	Weeks 1–3	25.00%	BD 340.909	Demonstration of Facility Master, Service Ticketing, Utility Logs & Lease Alerts
Milestone 2	Fixed Asset Master, QR Tagging, Custodian Tracking & Depreciation	Weeks 4–6	25.00%	BD 340.909	Sign-off of QR Asset Tagging, Scanner Integration, Transfers & Depreciation Engine
Milestone 3	Fleet Logistics, Vehicle Bookings, Fuel Logs & TCO Analytics	Weeks 7–9	25.00%	BD 340.909	Demonstration of Vehicle Master, Driver Bookings, Fuel Reconciliation & TCO Dashboard
Milestone 4	Staff Locker Control, Secure Digital Document Vault & Final UAT	Weeks 10–12	25.00%	BD 340.909	Delivery of Locker Registry, Digital Document Vault, CR Renewal Alerts & Final Multi-Branch Sign-off

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
Total Milestone Project Cost	Complete Module 6 ERP Scope Delivery	12 Weeks	100.00%	BD 1,022.727	Formal Acceptance & Sign-off



Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Working Calendar & Team Hours

Development operates on a 5-day standard work week (Sunday to Thursday, 09:00 to 18:00 AST), aligning with GCC financial business hours. Sprint demos are scheduled at the end of each 3-week milestone.

15-Day Milestone Review & Grace Period

Client is granted a formal 15-calendar-day review window following each sprint deployment to verify deliverables and audit tracking before milestone invoice closure.

SLA & Defect Severity Matrix

Critical Asset/Document Security Blockers (Severity 1) resolved within 4 hours; Major Functional Defects (Severity 2) within 24 hours; Minor UI adjustments (Severity 3) within 3 business days.

Scope Control & Document Security

Features adhere strictly to the 25-page specifications in ARCH-001. All legal document uploads and sensitive vehicle records are encrypted with AES-256 for corporate confidentiality.

Stakeholder Authorization & Sign-Off

By digitally endorsing this Milestone & Payment Structure document, all parties confirm adherence to the deliverables, 12-week timeline, and financial schedule defined for **SL-POP-ERP-MS-008**:

DEVELOPER / ARCHITECT

Ancy Varghese Thekkan
Software Architect, SaNDS Lab

✓ PREPARED (18-JUNE-2026)

COMPLETED

VERIFIED BY (DEVELOPER)

Ajit Kumar KV
CEO & Managing Director, SaNDS Lab

✓ VERIFIED (04-JULY-2026)

CHECKED & VERIFIED

CONSULTANT REVIEW

UniGlobal Business Solutions
Project Management Consultant

🕒 PENDING REVIEW

IN REVIEW

CLIENT APPROVAL

Popular Auto Spare & A/C Parts
Authorized Executive Signatory

🕒 PENDING APPROVAL

PENDING SIGNATURE